

Product Requirements Document

Gmail Commitment Radar

1. Product Summary

Product name

Gmail Commitment Radar

One-line description

Gmail Commitment Radar is an AI-assisted layer inside a Gmail-like inbox that detects promises, asks, deadlines, and pending follow-ups hidden inside email threads, then helps users track, resolve, snooze, or draft responses before commitments are missed.

Product thesis

Email productivity tools have focused heavily on writing faster, summarizing threads, and searching information. The deeper professional pain is that people lose track of what they promised, what others asked from them, and which threads require action. Gmail Commitment Radar turns email from a passive message archive into an active trust-preserving commitment system.

Core user problem

Users often miss commitments because they are buried inside long email threads, casually phrased, or spread across multiple conversations. Existing inbox affordances like stars, labels, snooze, and unread status are useful, but they require manual organization and do not understand the meaning of the conversation.

Product opportunity

A Gmail-native commitment layer can identify obligations automatically, show them in the same visual language as Gmail, and give users lightweight actions such as mark done, snooze, draft reply, add to calendar, or create task.

Target prototype goal

Build a high-fidelity Gmail-style prototype that feels like an extension of the real Gmail inbox. The product should not look like a separate dashboard pasted on top of Gmail. It should feel like a native Gmail side panel and inbox intelligence layer.

2. Problem Definition

Current state

Professionals use Gmail for recruiting, customer conversations, vendor coordination, manager updates, invoices, family logistics, interviews, project reviews, calendar scheduling, and personal follow-ups. Many of these conversations contain small but important commitments.

Examples:

- "I will send the deck tomorrow."
- "Can you send your updated resume before the call?"
- "Please confirm by Friday."
- "Let me check with finance and get back to you."
- "Can you share three times that work?"
- "I will review this by end of day."
- "Please send the signed form 24 hours before the meeting."
- "Following up here."
- "Just checking if you had a chance to look."

The problem is not that users cannot write a reply. The problem is that they often forget that a reply, file, confirmation, or decision is expected.

Why current inbox tools are insufficient

Stars

Stars help mark importance, but users must manually apply them. Stars also do not distinguish between "important someday" and "I owe someone something by tomorrow."

Labels

Labels help categorize messages, but they are not semantic. A thread labeled "Recruiting" still does not say whether the user owes availability, a resume, a portfolio link, or a response.

Snooze

Snooze is useful when the user remembers to snooze manually. It does not automatically understand when a promise exists.

Unread

Unread status is a weak proxy for action. A read email can still contain an unresolved commitment.

Search

Search helps users find something they remember. It does not surface commitments they forgot.

AI writing

AI writing helps create responses, but it does not maintain a durable commitment state across threads.

3. Product Vision

Gmail Commitment Radar should become the layer that answers four questions every time a user opens Gmail:

1. What do I owe other people?
2. What are other people waiting on from me?
3. What commitments are at risk today?
4. What is the fastest safe action I can take?

The product should feel calm, helpful, and non-judgmental. It should not create anxiety by turning every email into a task. It should only surface commitments with enough confidence and should clearly explain why each item was detected.

4. Target Users

Primary user: Busy professional

A person who receives many work-related emails and often has to respond to recruiters, hiring managers, vendors, teammates, customers, agencies, finance teams, legal teams, or executives.

Pain points

- Misses small follow-ups.
- Forgets to send documents before deadlines.
- Loses track of promises across many threads.
- Has to reread email chains to remember context.
- Feels inbox anxiety because important asks are mixed with notifications.

Success outcome

The user opens Gmail and immediately sees the most important commitments they owe today.

Secondary user: Product manager or cross-functional operator

A PM, program manager, founder, analyst, sales lead, recruiter, or account manager who coordinates work across many people.

Pain points

- Needs to follow up without sounding pushy.

- Needs to track who owes what.
- Has many threads with ambiguous ownership.
- Wants to protect credibility by responding on time.

Success outcome

The user can quickly identify action-required threads and send context-aware replies.

Tertiary user: Personal life organizer

A user managing appointments, documents, payments, school forms, family logistics, travel plans, or household tasks through email.

Pain points

- Misses administrative tasks.
- Forgets appointment preparation.
- Loses deadlines in long confirmation emails.

Success outcome

The user gets clear, lightweight reminders without manually creating tasks.

5. Jobs To Be Done

JTBD 1: Detect my obligations

When I receive or send email, I want Gmail to identify whether I owe someone a response, document, confirmation, decision, payment, review, or scheduling action so that I do not forget my responsibilities.

JTBD 2: Understand urgency

When an email contains a commitment, I want to know whether it is due today, later, overdue, or low risk so that I can prioritize properly.

JTBD 3: Take action quickly

When a commitment is detected, I want one-click actions like draft reply, mark done, snooze, add to calendar, create task, or open thread so that I can resolve it without switching tools.

JTBD 4: Trust the detection

When AI marks something as a commitment, I want to see the exact phrase and reasoning so that I can verify it quickly.

JTBD 5: Avoid noise

When I receive newsletters, promotions, automated notifications, or FYI emails, I do not want them turned into commitments unless there is a clear action required from me.

6. Product Principles

Native to Gmail

The feature should look like it belongs inside Gmail. It should use Gmail-style layout, spacing, typography, iconography, pill shapes, panels, hover states, and muted colors.

Calm, not alarmist

Use urgency language carefully. Avoid aggressive warnings. Prefer “Due today,” “At risk,” and “Waiting on you” over fear-driven language.

Explainable AI

Every detected commitment must show the source phrase, inferred owner, due date, and confidence level.

Human override first

Users must be able to correct AI mistakes quickly. Actions like “Not a commitment,” “Wrong owner,” “Wrong date,” and “Already done” are required.

Lightweight actions

The product should reduce work, not create another project management system.

Privacy respectful

The prototype should demonstrate privacy by design. It should not imply that emails are sent to unknown third parties without user consent.

High signal over coverage

It is better to miss some low-confidence commitments than to overwhelm the user with noisy tasks.

7. Scope

MVP scope

The MVP should include:

1. Gmail-style inbox shell.
2. Left Gmail navigation.
3. Top Gmail search and toolbar.
4. Message list with commitment indicators.
5. Thread view.
6. Right-side Commitment Radar panel.
7. Commitment extraction from seeded sample emails.
8. Commitment detail cards.
9. Risk scoring.
10. Action buttons:
11. Mark done.
12. Snooze.
13. Draft reply.
14. Add to Calendar.
15. Create Task.
16. Not a commitment.
17. Commitment dashboard view.
18. Confidence and source phrase explanation.
19. Basic settings for commitment detection.
20. Empty states and error states.
21. Local mock data persistence.

V1 scope

V1 should include:

1. Multi-thread commitment aggregation.
2. Natural language search across commitments.
3. Commitment ownership classification:
4. I owe.
5. They owe.
6. Mutual.
7. Delegated.
8. Unknown.
9. Follow-up suggestion generation.

10. Gmail label integration simulation.
11. Calendar integration simulation.
12. Google Tasks integration simulation.
13. User correction learning.
14. Contact importance weighting.
15. Notification preferences.
16. Due-date parsing from relative dates.
17. Cross-thread duplicate detection.

Out of scope for MVP

1. Real Gmail API integration.
 2. Real email sending.
 3. Real Google Calendar writes.
 4. Real Google Tasks writes.
 5. Background email scanning.
 6. Multi-account support.
 7. Enterprise admin controls.
 8. Mobile app.
 9. Voice interface.
 10. Attachments upload or download.
 11. Real notification delivery.
 12. Permanent cloud storage.
 13. Full OAuth implementation.
-

8. Gmail UI Matching Requirements

Overall visual direction

The interface should look like modern Gmail desktop. The prototype should use a light, spacious, Material-style UI with rounded surfaces, soft blue-gray backgrounds, Google Sans-style typography, icon-only utility rails, and clear separation between inbox list, thread content, and side panel.

Layout structure

The app must use a four-zone Gmail-like layout:

1. Top app bar.
2. Left navigation sidebar.
3. Main content area.
4. Right utility rail or commitment side panel.

Top app bar

The top app bar should span the full width of the viewport.

Required elements from left to right:

1. Hamburger menu icon.
2. Gmail-style logo area.
3. Search input.
4. Search filter icon.
5. Help icon.
6. Settings gear icon.
7. Google apps grid icon.
8. User avatar.

Search input requirements:

- Placeholder: "Search mail".
- Shape: rounded pill.
- Height: approximately 48px.
- Background: soft blue-gray.
- Width: flexible, centered, max width around 720px.
- Leading search icon.
- Optional trailing tune/filter icon.
- On focus, background should become white.
- On focus, a subtle shadow should appear.
- Keyboard shortcut hint can be shown on hover or inside search suggestions.

Left navigation sidebar

The left sidebar should feel Gmail-native.

Required items:

1. Compose button.
2. Inbox.
3. Starred.
4. Snoozed.
5. Sent.
6. Drafts.
7. More.
8. Labels section.
9. Commitment Radar entry.

Compose button requirements:

- Large rounded pill.
- Light blue fill.
- Pencil or plus icon.
- Label: "Compose".
- Width around 140px.
- Height around 56px.

- Hover state slightly darker.
- Shadow minimal.

Commitment Radar navigation entry:

- Should appear as a product extension inside the left nav.
- Label: "Commitment Radar".
- Icon: radar, target, check-circle, or assistant sparkle.
- Badge count: number of open commitments.
- Selected state should use Gmail-style light blue background.
- Badge should use subtle red or amber only when there are urgent commitments.
- Should not overpower Inbox.

Left nav selected state:

- Rounded right pill.
- Background: light blue.
- Font weight: medium.
- Icon and text color: dark gray or Gmail blue.
- Count aligned right.

Main content area

Main content must support two modes:

1. Inbox list mode.
2. Thread detail mode.

The main content area should use Gmail-like cards and rows rather than a heavy dashboard style.

Right side panel

The product should use the right side as the primary home for Commitment Radar.

There are two acceptable panel patterns:

Pattern A: Collapsed utility rail plus expandable panel.

- Default right rail width: around 56px.
- Icons stacked vertically.
- Commitment Radar icon appears alongside Calendar, Keep, Tasks-style utilities.
- Clicking the Radar icon opens a full-height side panel.

Pattern B: Always-visible side panel.

- Width: 360px to 420px.
- Appears only when Commitment Radar is active.
- Does not cover the email thread.

- Main email area shrinks responsively.

Preferred MVP approach:

Use Pattern B for demo clarity, but include a collapsed rail state in design.

Typography

Use the following type hierarchy:

App shell

- Font family: Google Sans-like, Arial fallback.
- Base size: 14px.
- Body color: #202124.
- Secondary color: #5f6368.

Inbox rows

- Sender: 14px, medium weight for unread.
- Subject: 14px, medium for unread, regular for read.
- Snippet: 14px, #5f6368.
- Timestamp: 12px, medium for unread.
- Badges: 11px to 12px.

Panel titles

- Section title: 16px, medium.
- Card title: 14px, medium.
- Metadata: 12px.
- Explanation text: 13px.

Color tokens

Use Gmail-inspired prototype tokens:

- App background: #f6f8fc.
- Main surface: #ffffff.
- Search background: #eaf1fb.
- Selected nav background: #d3e3fd.
- Compose background: #c2e7ff.
- Primary text: #202124.
- Secondary text: #5f6368.
- Divider: #e0e0e0.
- Hover background: #f2f6fc.
- Active blue: #1a73e8.
- Low risk green: #188038.
- Medium risk amber: #f9ab00.

- High risk red: #d93025.
- AI accent purple or blue: #5e35b1 or #1a73e8, used sparingly.

Iconography

Use Material Symbols or a visually similar icon set.

Required icons:

- Menu.
- Search.
- Tune or filter.
- Help.
- Settings.
- Apps grid.
- Compose.
- Inbox.
- Star.
- Snooze.
- Send.
- Draft.
- Check circle.
- Schedule.
- Calendar.
- Reply.
- More vertical.
- Open in new.
- Close.
- Radar or target.
- Sparkle for AI explanation.
- Warning or priority indicator.

Icon sizes:

- Top bar icons: 24px.
- Nav icons: 20px to 24px.
- Inbox row hover icons: 20px.
- Commitment card action icons: 18px.
- Utility rail icons: 24px.

Spacing

Use 8px grid spacing.

Recommended dimensions:

- Top app bar height: 64px.
- Left nav expanded width: 256px.

- Left nav collapsed width: 72px.
- Utility rail width: 56px.
- Commitment panel width: 380px.
- Inbox row height: 48px to 56px.
- Thread toolbar height: 48px.
- Card border radius: 12px to 16px.
- Search input height: 48px.
- Compose button height: 56px.

Interaction states

Every clickable element must include:

1. Default state.
2. Hover state.
3. Focus-visible state.
4. Active or pressed state.
5. Disabled state when applicable.

Focus states must be keyboard accessible and visible.

9. Core Product Surfaces

9.1 Inbox List View

Objective

Help users see which email threads contain unresolved commitments without leaving the normal inbox list.

Layout

Each inbox row should include:

1. Checkbox.
2. Star.
3. Sender.
4. Subject.
5. Snippet.
6. Commitment badge when applicable.
7. Attachment icon if applicable.
8. Timestamp.
9. Hover quick actions.

Commitment badge types

Badge: "You owe"

Used when the user has a detected obligation.

Visual:

- Rounded pill.
- Light amber or light red background depending on urgency.
- Text: "You owe".
- Optional due date: "Today" or "Fri".

Example:

"You owe · Today"

Badge: "They owe"

Used when another person owes the user a response or action.

Visual:

- Rounded pill.
- Light blue background.
- Text: "They owe".

Badge: "Waiting"

Used when the thread is waiting for someone else and the user does not need to act yet.

Visual:

- Light gray or blue-gray pill.
- Text: "Waiting".

Badge: "Done"

Used when a commitment was detected and resolved.

Visual:

- Light green pill.
- Text: "Done".

Badge: "Review"

Used when confidence is medium and the user should confirm.

Visual:

- Light purple or neutral pill.
- Text: "Review".

Row priority indicators

For high-risk commitments:

- A subtle left border or small red dot can appear near the sender.
- Avoid aggressive warning banners inside the inbox row.
- The row should still look like Gmail.

Inbox row hover actions

On hover, replace timestamp with quick actions:

1. Archive.
2. Delete.
3. Mark unread.
4. Snooze.
5. Radar quick action.

Radar quick action opens a mini popover:

- Commitment summary.
- Due date.
- Primary action.
- "Open Radar panel".

Inbox tabs

The prototype may include Gmail-style category tabs:

1. Primary.
2. Promotions.
3. Social.
4. Updates.

Commitment Radar should primarily operate on Primary and Updates. Promotions and Social should be ignored unless manually included.

Acceptance criteria

- User can scan inbox and identify at least five threads with different commitment states.
- Hovering a thread reveals Gmail-like quick actions.
- Clicking a commitment badge opens the side panel with the selected commitment.
- Clicking the row opens the thread detail view.

- The UI remains readable at 1366px width.
-

9.2 Thread Detail View

Objective

Show commitment intelligence in context while the user reads an email thread.

Layout

Thread view should include:

1. Gmail-style top thread toolbar.
2. Email subject.
3. Message cards.
4. Sender details.
5. Reply area.
6. Right Commitment Radar panel.

Thread toolbar

Required actions:

- Back.
- Archive.
- Report spam.
- Delete.
- Mark unread.
- Snooze.
- Add to tasks.
- Move.
- Labels.
- More.

Commitment highlight inside email body

Detected source phrases should be highlighted subtly.

Example highlight:

"I will send the updated deck by Friday."

Visual style:

- Soft yellow or light blue underline.
- No heavy highlighter effect.

- Hover shows tooltip:
- "Detected commitment"
- Owner: "You"
- Due: "Friday"
- Confidence: "92 percent"

Thread-level commitment summary

At the top of the thread, below the subject, show a small native Gmail-like summary chip only if commitment exists.

Example:

"1 open commitment · Due today"

Clicking it scrolls the side panel to the relevant card.

Reply box integration

When the user clicks "Draft reply," Gmail Commitment Radar should insert a draft into a Gmail-like reply composer.

The draft should be editable. It should never auto-send.

Reply composer requirements:

- Looks like Gmail inline reply.
- Includes To field only if expanded.
- Body text area.
- Formatting toolbar optional.
- Send button in Gmail blue.
- More options menu.
- Trash draft icon.

Acceptance criteria

- User can open a thread and see the exact source phrase that produced the commitment.
- User can distinguish thread content from AI overlay.
- User can draft a reply from a commitment card.
- User can mark the commitment done without leaving the thread.

9.3 Commitment Radar Side Panel

Objective

Provide the main intelligence and action surface for commitments.

Panel structure

The side panel should include:

1. Header.
2. Today summary.
3. Filter chips.
4. Commitment cards.
5. Explanation drawer.
6. Action area.
7. Settings entry.

Header

Header content:

- Icon: Radar or target.
- Title: "Commitment Radar".
- Subtitle: "3 need attention today."
- Close icon.
- More menu.

Optional top status:

- "Last scanned just now."
- "AI suggestions are private to this prototype."

Today summary

A compact summary card at the top:

- "Today"
- Count of due commitments.
- Count of overdue commitments.
- Count of waiting-on-others items.

Example:

"Today: 3 open · 1 overdue · 2 waiting"

Filter chips

Required filters:

1. All.
2. You owe.
3. They owe.
4. Due today.

5. Overdue.
6. Review.
7. Done.

Chip behavior:

- Selected chip uses light blue background.
- Multi-select is not needed in MVP.
- Count can appear inside chip.

Commitment card

Each commitment card must include:

1. Status badge.
2. Commitment title.
3. Thread source.
4. Contact or sender.
5. Due date.
6. Risk level.
7. Confidence.
8. Source phrase.
9. Explanation.
10. Primary action.
11. Secondary actions.

Card example

Status: You owe

Title: Send updated resume

Thread: Interview preparation call

Contact: James

Due: 24 hours before call

Risk: High

Confidence: 91 percent

Source phrase: "Please send your resume or updated LinkedIn URL at least 24 hours prior to our call."

Primary action: Draft reply

Secondary actions: Mark done, Snooze, Not a commitment

Card density modes

The panel should support:

Compact mode

- Shows title, due date, status, primary action.
- Used when many commitments exist.

Expanded mode

- Shows source phrase, explanation, and all actions.
- Used when a card is selected.

MVP can default to expanded for selected thread and compact for all others.

Risk levels

Low

Criteria:

- No explicit deadline.
- Soft ask.
- Not from high-importance contact.
- User has not promised action.

Text:

"Low risk"

Visual:

Green or neutral gray.

Medium

Criteria:

- Clear ask but no immediate deadline.
- Due in more than 24 hours.
- Medium confidence.

Text:

"Medium risk"

Visual:

Amber.

High

Criteria:

- Due today or overdue.
- User explicitly promised action.

- Thread has follow-up language.
- Sender is important.
- Deadline includes “mandatory,” “before call,” “by EOD,” or similar language.

Text:

“High risk”

Visual:

Red or strong amber.

Primary actions by commitment type

User owes response

Primary action: “Draft reply”

Secondary actions:

- Mark done.
- Snooze.
- Add to task.
- Not a commitment.

User owes document

Primary action: “Draft reply”

Secondary actions:

- Attach file, prototype-only disabled state.
- Mark done.
- Snooze.
- Add to task.

User owes scheduling response

Primary action: “Send availability”

Secondary actions:

- Open calendar, simulated.
- Draft reply.
- Snooze.

User owes review

Primary action: "Create task"

Secondary actions:

- Draft reply.
- Mark done.
- Snooze.

Other person owes user

Primary action: "Draft follow-up"

Secondary actions:

- Keep waiting.
- Snooze.
- Mark resolved.

Ambiguous ownership

Primary action: "Review"

Secondary actions:

- Assign to me.
- Assign to sender.
- Not a commitment.

Explanation drawer

Every commitment card should include a collapsible "Why am I seeing this?" drawer.

Required fields:

- Detected phrase.
- Parsed action.
- Parsed owner.
- Parsed due date.
- Confidence score.
- Signals used.
- User override controls.

Example:

"Detected because the sender asked for an updated LinkedIn URL and used a deadline phrase: at least 24 hours prior."

Acceptance criteria

- Panel opens from right side.
 - Panel can be closed.
 - Panel can filter commitments.
 - Cards show source phrase and confidence.
 - User can take action from card.
 - Actions update the commitment state immediately.
-

9.4 Commitment Dashboard View

Objective

Provide a full inbox-level view of all commitments.

Entry points

User can open dashboard from:

1. Left nav "Commitment Radar".
2. Right panel "View all".
3. Search query suggestions.
4. Empty inbox card.

Dashboard layout

Dashboard should look Gmail-native, not like a separate SaaS tool.

Top section:

- Page title: "Commitment Radar".
- Subtitle: "Track promises, asks, and follow-ups across your inbox."
- Search commitments input.
- Filter chips.

Summary cards:

1. Due today.
2. Overdue.
3. Waiting on others.
4. Needs review.

List section:

- Commitment rows.
- Sort options.

- Bulk actions.

Commitment row fields

1. Checkbox.
2. Status icon.
3. Commitment title.
4. Thread subject.
5. Contact.
6. Due date.
7. Risk.
8. Last email date.
9. Primary action.
10. More menu.

Sort options

1. Risk.
2. Due date.
3. Newest.
4. Contact importance.
5. Confidence.

Bulk actions

1. Mark done.
2. Snooze.
3. Add to tasks.
4. Dismiss.
5. Export, disabled in MVP.

Acceptance criteria

- User can see all open commitments.
- User can filter by owner and due date.
- User can select multiple commitments.
- User can open the original thread from any row.

10. Commitment Types

Type 1: Promise made by user

Definition:

The user sent or implied that they would do something.

Examples:

- "I will send this tomorrow."
- "Let me get back to you."
- "I can share the document by Friday."
- "I will confirm after checking."

Owner:

User.

Default status:

Open.

Primary action:

Mark done or draft update.

Type 2: Ask from someone else

Definition:

Someone requested an action from the user.

Examples:

- "Can you send your resume?"
- "Please confirm your availability."
- "Could you review this?"
- "Please sign the attached document."

Owner:

User.

Default status:

Open.

Primary action:

Draft reply.

Type 3: Waiting on someone else

Definition:

The user asked someone else for something and is waiting.

Examples:

- "Can you send the contract?"
- "Please confirm the timeline."
- "Following up on the invoice."

Owner:

Other person.

Default status:

Waiting.

Primary action:

Draft follow-up.

Type 4: Meeting preparation requirement

Definition:

An email contains action required before a meeting.

Examples:

- "Please send your resume 24 hours prior."
- "Review the attached agenda before the call."
- "Complete the form before your appointment."

Owner:

Usually user.

Default status:

Open.

Primary action:

Create task or draft reply.

Type 5: Deadline confirmation

Definition:

The user must confirm, approve, pay, submit, or decide by a deadline.

Examples:

- "Please confirm by Friday."
- "Payment is due tomorrow."
- "Approval needed by EOD."

Owner:

User.

Default status:

Open.

Primary action:

Draft reply or create task.

Type 6: Soft follow-up risk

Definition:

No explicit deadline exists, but thread context suggests response is socially or professionally expected.

Examples:

- "Just checking in."
- "Following up here."
- "Wanted to see if you had a chance."

Owner:

User or other person depending on previous email.

Default status:

Review.

Primary action:

Review or draft reply.

11. AI Detection Requirements

Extraction pipeline

The prototype should simulate the following pipeline:

1. Load email thread.
2. Segment messages.
3. Identify sender and recipient roles.
4. Extract action phrases.
5. Classify commitment type.
6. Determine owner.
7. Parse due date.
8. Score risk.
9. Assign confidence.
10. Generate explanation.
11. Render commitment card.

Required extracted fields

Each commitment object must include:

- commitment_id
- thread_id
- message_id
- source_message_index
- source_phrase
- normalized_action
- commitment_type
- owner_type
- owner_name
- recipient_name
- due_date
- due_date_text
- due_date_confidence
- risk_level
- confidence_score
- status
- created_at
- updated_at
- resolved_at
- snoozed_until

- explanation
- suggested_action
- draft_reply

Owner classification

Owner types:

1. me
2. other_person
3. shared
4. unknown

Rules:

- If user says “I will,” owner is me.
- If sender says “please send,” owner is me.
- If user asks “can you send,” owner is other_person.
- If a group is asked, owner is shared.
- If thread context is unclear, owner is unknown and card status is Review.

Due date parsing

Supported MVP date formats:

1. Today.
2. Tomorrow.
3. Friday.
4. End of day.
5. EOD.
6. End of week.
7. 24 hours before call.
8. Before our meeting.
9. By next week.
10. No explicit date.

For MVP, relative dates can be simulated using a fixed current date.

Confidence scoring

Confidence score should range from 0 to 100.

High confidence: 85 to 100

Used when:

- Clear action verb exists.
- Owner is clear.

- Due date is explicit or implied by meeting.
- Source phrase is direct.

Medium confidence: 60 to 84

Used when:

- Action exists but owner or deadline is ambiguous.
- Language is polite or indirect.
- Thread context helps but does not fully resolve.

Low confidence: below 60

Used when:

- Phrase could be FYI.
- No clear owner.
- No clear action.
- Auto-generated email may contain generic wording.

MVP should not create open commitments below 60 confidence. Low-confidence items can appear in "Review" only if useful.

Risk scoring formula

Risk should be calculated using a simple weighted model.

Risk score inputs:

- Due date proximity.
- Overdue status.
- Commitment type.
- Contact importance.
- Thread recency.
- Follow-up language.
- User promise language.
- Confidence score.

Example scoring:

- Overdue: +40.
- Due today: +30.
- Due within 24 hours: +25.
- Explicit user promise: +20.
- Sender asks directly: +15.
- Important contact: +15.
- Follow-up phrase: +15.
- Has meeting preparation requirement: +10.

- Confidence below 75: -10.

Risk levels:

- 0 to 34: Low.
- 35 to 69: Medium.
- 70 to 100: High.

AI explanation requirement

Every AI-generated card must explain itself in plain language.

Good explanation:

“This was detected because the sender asked you to send an updated resume and included a deadline phrase: at least 24 hours prior.”

Bad explanation:

“Model classified this as type ask with owner me and confidence 0.91.”

Hallucination prevention

The system must not invent commitments that are not grounded in email text.

Requirements:

- Every commitment must include source phrase.
- Source phrase must be visible to the user.
- Draft replies must not claim completion unless user marks done.
- Draft replies must not invent attachments.
- Draft replies must not invent availability.
- Draft replies must not say “attached” unless attachment exists.
- Draft replies must not mention private information outside the thread.

12. User Actions

Mark done

Behavior:

- Changes status to Done.
- Removes item from active list.
- Keeps item visible in Done filter.
- Shows undo snackbar.

Snackbar:

"Commitment marked done. Undo"

Snooze

Behavior:

- Opens Gmail-style snooze menu.
- User can select:
 - Later today.
 - Tomorrow.
 - This weekend.
 - Next week.
 - Pick date and time.
- Commitment status becomes Snoozed.
- It reappears when snooze time arrives.

MVP:

Snooze can update state locally without real notifications.

Draft reply

Behavior:

- Generates a draft response based on commitment type.
- Opens Gmail-style inline reply composer.
- User can edit.
- Send button is disabled or simulated in MVP.

Draft principles:

- Short.
- Professional.
- Grounded in thread.
- Does not overpromise.
- Includes uncertainty when needed.

Example for sending document later:

"Hi James, thanks for the note. I will send over my updated resume before the deadline."

Example for follow-up:

"Hi Sarah, just following up on this. Please let me know when you have a chance."

Add to Calendar

Behavior:

- For meeting-related commitments, opens a simulated Google Calendar modal.
- Pre-fills title, date, and reminder.
- Does not create a real calendar event in MVP.

Create Task

Behavior:

- Opens a Google Tasks-style mini modal.
- Pre-fills task title and due date.
- Adds task to local mock list.

Not a commitment

Behavior:

- Marks the item as dismissed.
- Asks optional feedback:
- Not an action.
- Wrong owner.
- Wrong date.
- Already handled.
- Other.
- Uses feedback to update local detection explanation in prototype.

Assign owner

For ambiguous commitments:

- Assign to me.
- Assign to sender.
- Assign to someone else.
- Shared.

Edit due date

User can manually edit due date.

Required behavior:

- Click due date.
- Open date picker.
- Save.
- Show "Updated by you."

Acceptance criteria

- Every action changes visible UI state.
 - Every action has an undo path when destructive.
 - User correction is visible in the card.
 - Draft reply is always editable.
-

13. Settings Requirements

Entry points

Settings can be opened from:

1. Commitment Radar panel more menu.
2. Dashboard settings button.
3. Gmail settings gear simulation.

Settings sections

Detection

Options:

- Detect commitments from received emails.
- Detect promises from sent emails.
- Detect waiting-on-others threads.
- Detect meeting preparation tasks.
- Ignore promotions and newsletters.
- Ignore automated notifications.

Risk

Options:

- Treat direct asks as high priority.
- Treat recruiters and hiring contacts as high priority.
- Treat finance, legal, and medical senders as high priority.
- Show low-confidence items in Review.

Notifications

Options:

- Show badge in left navigation.
- Show inbox row badges.
- Show daily summary.

- Show overdue banner.
- Snooze reminders.

Privacy

Options:

- Show source phrase for every AI suggestion.
- Clear local AI suggestions.
- Disable Commitment Radar.

MVP settings behavior

Settings should update prototype behavior locally where practical.

At minimum:

- Toggle inbox badges on and off.
 - Toggle low-confidence review items.
 - Toggle automated email filtering.
 - Clear completed commitments.
-

14. Data Model

Thread

Fields:

- id
- subject
- participants
- last_message_at
- unread
- starred
- labels
- category
- has_attachment
- messages
- commitments

Message

Fields:

- id
- thread_id

- sender_name
- sender_email
- recipients
- sent_at
- body
- snippet
- is_from_user
- attachments
- detected_phrases

Commitment

Fields:

- id
- thread_id
- message_id
- title
- type
- owner_type
- owner_name
- source_phrase
- normalized_action
- due_date
- due_date_text
- confidence
- risk
- status
- created_at
- updated_at
- snoozed_until
- resolved_at
- suggested_actions
- draft_reply
- explanation
- user_overrides

User override

Fields:

- commitment_id
- override_type
- previous_value
- new_value
- created_at

Suggested action

Fields:

- id
 - commitment_id
 - type
 - label
 - primary
 - enabled
 - disabled_reason
-

15. Sample Seed Data Requirements

The prototype should include 10 to 12 seeded threads.

Thread 1: Recruiting preparation

Commitment:

Send updated resume or LinkedIn URL 24 hours before call.

Type:

Meeting preparation requirement.

Risk:

High.

Primary action:

Draft reply.

Thread 2: Manager follow-up

Commitment:

Send roadmap summary by Friday.

Type:

Promise made by user.

Risk:

Medium or high depending on current date.

Primary action:

Draft update.

Thread 3: Vendor invoice

Commitment:

Approve invoice by end of week.

Type:

Deadline confirmation.

Risk:

Medium.

Primary action:

Create task.

Thread 4: Customer feedback

Commitment:

Review feedback document.

Type:

Ask from someone else.

Risk:

Medium.

Primary action:

Create task.

Thread 5: Friend travel plan

Commitment:

Confirm weekend travel timing.

Type:

Personal logistics.

Risk:

Low.

Primary action:

Draft reply.

Thread 6: Waiting for contract

Commitment:

Other person owes signed agreement.

Type:

Waiting on someone else.

Risk:

Low.

Primary action:

Draft follow-up.

Thread 7: Medical appointment

Commitment:

Complete intake form before appointment.

Type:

Meeting preparation requirement.

Risk:

High.

Primary action:

Create task.

Thread 8: Automated newsletter

Commitment:

None.

Type:

No commitment.

Risk:

None.

Expected behavior:

No badge.

Thread 9: Promotion

Commitment:

None.

Type:

Ignored.

Risk:

None.

Expected behavior:

No badge.

Thread 10: Ambiguous follow-up

Commitment:

Possible response needed.

Type:

Soft follow-up risk.

Risk:

Review.

Primary action:

Review.

16. Search Requirements

Gmail search compatibility

The main search bar remains "Search mail."

Commitment Radar should add suggested search chips when active.

Examples:

- "commitments due today"
- "things I owe"
- "waiting on others"
- "overdue commitments"
- "resume before call"
- "threads where I promised to follow up"

Search results behavior

When the query is commitment-related:

- Show normal email results.
- Also show a Commitment Radar result module at top.
- Module includes matching commitments.
- Clicking a commitment opens thread and side panel.

Dashboard search

Dashboard search should search across:

- Commitment title.
 - Source phrase.
 - Thread subject.
 - Sender.
 - Due date text.
 - Status.
-

17. Empty States

No commitments detected

Message:

"You are all caught up."

Subtext:

"Commitment Radar did not find open promises, asks, or follow-ups in this view."

Actions:

- View done.
- Adjust settings.

No commitments in selected thread

Message:

"No commitments found in this thread."

Subtext:

"Radar looks for promises, asks, deadlines, and follow-ups."

Action:

- Add manually.

Low confidence only

Message:

"Nothing urgent found."

Subtext:

"There are a few items Radar is unsure about."

Action:

- Review suggestions.

Detection disabled

Message:

"Commitment Radar is off."

Subtext:

"Turn it on to detect promises and follow-ups in your inbox."

Action:

- Turn on.
-

18. Error States

AI detection unavailable

Message:

"Radar could not analyze this thread."

Subtext:

"Try again or add a commitment manually."

Action:

- Retry.

Draft generation failed

Message:

"Draft could not be generated."

Subtext:

"You can still write a reply manually."

Action:

- Try again.

No source phrase found

This should not happen for AI-generated commitments. If it does:

Message:

"This suggestion needs review."

Subtext:

"Radar could not find the exact source phrase."

Action:

- Dismiss.

Local storage full

Message:

"Changes could not be saved locally."

Subtext:

"Your browser storage may be full."

Action:

- Clear demo data.

19. Accessibility Requirements

Keyboard navigation

User must be able to:

- Navigate inbox rows.

- Open thread.
- Open Commitment Radar panel.
- Move between commitment cards.
- Trigger primary actions.
- Close modals.
- Use filters.
- Return focus after closing panel.

Focus management

- Opening panel moves focus to panel header.
- Closing panel returns focus to triggering button.
- Opening modal traps focus.
- Closing modal returns focus to action button.

Screen reader labels

Required labels:

- "Open Commitment Radar"
- "Commitment due today"
- "High risk commitment"
- "Mark commitment done"
- "Snooze commitment"
- "Draft reply for commitment"
- "Source phrase"
- "Confidence score"

Color contrast

- Text must meet WCAG AA contrast.
- Risk status must not rely on color alone.
- Use text labels with color indicators.

Reduced motion

- Respect prefers-reduced-motion.
- Panel animations should be disabled or reduced.

20. Responsive Requirements

Desktop first

Primary target:

- 1366px by 768px.
- 1440px by 900px.

- 1920px by 1080px.

Tablet

At widths below 1024px:

- Left nav collapses.
- Commitment panel becomes overlay.
- Inbox remains primary.

Mobile

Mobile is not required for MVP.

However, layout should not break below 768px. It can show a message:

“Commitment Radar prototype is optimized for desktop Gmail.”

21. Performance Requirements

Initial load

- Prototype should load within 2 seconds on local machine.
- Mock data should render instantly.

Interaction latency

- Opening panel: under 150ms.
- Filtering commitments: under 100ms.
- Mark done: under 100ms.
- Draft generation simulation: under 800ms with loading state.

Loading states

Use skeletons or subtle progress indicators.

Do not use full-page spinners except on initial app load.

22. Technical Requirements

Recommended stack

- Next.js or Vite React.
- TypeScript.
- Tailwind CSS.

- Local mock data.
- LocalStorage or IndexedDB for state persistence.
- Optional Zustand or React Context for state.
- Material Symbols or Lucide icons styled to feel Gmail-like.

Component architecture

Recommended components:

- AppShell
- TopBar
- GmailSearchBar
- LeftNav
- InboxList
- InboxRow
- ThreadView
- ThreadToolbar
- MessageCard
- CommitmentBadge
- CommitmentPanel
- CommitmentCard
- CommitmentExplanation
- CommitmentDashboard
- FilterChips
- SnoozeMenu
- DraftReplyComposer
- SettingsModal
- EmptyState
- SnackBar

State management

State should include:

- selectedThreadId
- selectedCommitmentId
- activeView
- panelOpen
- activeFilter
- searchQuery
- threads
- commitments
- settings
- snackbar

Mock AI layer

Create a deterministic local function:

analyzeThread(thread): Commitment[]

This function should use seeded examples and rule-based phrase matching. It should simulate AI output with confidence and explanation.

Do not call a real model in MVP unless optional.

Optional AI integration

If using an LLM later, the model prompt must require JSON output with:

- source_phrase
- action
- owner
- due_date_text
- confidence
- explanation
- suggested_reply

The system must reject outputs without source phrase grounding.

23. Manual Add Commitment

Objective

Allow user to create a commitment when AI misses one.

Entry points

1. Thread panel.
2. Dashboard.
3. Inbox row more menu.

Fields

- Title.
- Owner.
- Due date.
- Source thread.
- Notes.
- Risk.
- Status.

Acceptance criteria

- User can add commitment manually.

- Manual commitments appear in panel and dashboard.
 - Manual commitments show “Added by you” instead of AI confidence.
-

24. Metrics

Activation metrics

- Percent of users who open Commitment Radar panel.
- Percent of detected commitments viewed.
- Percent of users who take at least one action.
- Time to first action.

Engagement metrics

- Commitments marked done per week.
- Draft replies generated.
- Snoozes created.
- Dashboard opens.
- Filter usage.

Quality metrics

- Detection acceptance rate.
- Dismissal rate.
- Wrong owner correction rate.
- Wrong date correction rate.
- Draft edit rate.
- False positive rate.
- False negative reports.

Outcome metrics

- Reduction in overdue commitments.
- Faster response time for action-required emails.
- Increase in user trust score.
- Weekly retained users.
- User-reported inbox confidence.

North star metric

Weekly resolved commitments per active user.

Guardrail metrics

- Dismissal rate does not exceed acceptable threshold.
- User does not disable feature after first session.
- Draft reply error rate remains low.

- Low-confidence suggestions do not dominate panel.
 - No private or unsupported claim appears in draft replies.
-

25. Product Analytics Events

Panel events

- radar_panel_opened
- radar_panel_closed
- radar_filter_selected
- radar_card_expanded
- radar_explanation_opened

Commitment events

- commitment_detected
- commitment_viewed
- commitment_marked_done
- commitment_snoozed
- commitment_dismissed
- commitment_due_date_edited
- commitment_owner_changed
- commitment_added_manually

Draft events

- draft_reply_generated
- draft_reply_inserted
- draft_reply_edited
- draft_reply_discarded
- draft_reply_simulated_send_clicked

Dashboard events

- radar_dashboard_opened
- commitment_search_performed
- commitment_bulk_action_used

Settings events

- radar_setting_changed
 - radar_disabled
 - radar_enabled
-

26. Privacy and Trust Requirements

Source grounding

Every AI suggestion must be grounded in the email thread.

Required:

- Show source phrase.
- Show confidence.
- Show owner.
- Show due date.
- Show explanation.

User control

User can:

- Dismiss suggestion.
- Edit owner.
- Edit due date.
- Mark done.
- Disable detection.
- Clear local suggestions.

Prototype privacy copy

Include subtle copy in settings:

“This prototype uses seeded local email data. No real emails are sent or scanned.”

If later connected to real Gmail:

“Commitment Radar analyzes selected email content to detect promises, asks, and follow-ups. You can turn it off at any time.”

Sensitive content handling

The prototype should avoid deeply sensitive seeded content. Do not include medical details, legal issues, financial account numbers, or private personal information beyond generic examples.

27. Demo Storyboard

Demo flow 1: Inbox scan

1. User opens Gmail-style inbox.
2. Commitment badges appear on several threads.
3. User sees “You owe · Today” on one thread.
4. User clicks the badge.
5. Right panel opens.
6. Commitment card explains the detected ask.

Demo flow 2: Thread action

1. User opens the thread.
2. Source phrase is highlighted inside email body.
3. Panel shows high-risk commitment.
4. User clicks “Draft reply.”
5. Inline Gmail-style composer opens.
6. Draft is inserted.
7. User edits draft.
8. User marks commitment done.
9. Snackbar appears with undo.

Demo flow 3: Dashboard

1. User clicks “Commitment Radar” in left nav.
2. Dashboard opens.
3. User filters by “Due today.”
4. User sees all commitments due today.
5. User snoozes one and marks another done.

Demo flow 4: AI mistake correction

1. User opens a Review commitment.
2. User sees ambiguous ownership.
3. User clicks “Not a commitment.”
4. Feedback menu appears.
5. User selects “Not an action.”
6. Item disappears from active list.

28. Acceptance Criteria Summary

Must have

- Gmail-like shell.
- Gmail-like search bar, nav, inbox rows, and right panel.

- Commitment badges in inbox.
- Commitment side panel.
- Commitment cards with source phrase, confidence, risk, and actions.
- Thread view with source phrase highlight.
- Draft reply generation simulation.
- Mark done, snooze, dismiss, and review actions.
- Commitment dashboard.
- Settings modal.
- Local state persistence.
- Seeded demo data.
- Empty and error states.

Should have

- Collapsed right rail.
- Dashboard bulk actions.
- Manual add commitment.
- User correction feedback.
- Search across commitments.
- Risk scoring explanation.

Could have

- Animated radar scan.
- Daily digest card.
- Contact importance scoring.
- Calendar and Tasks mock integrations.
- Dark mode.
- Mobile layout.

Will not have in MVP

- Real Gmail API.
- Real email sending.
- Real Google Calendar writes.
- Real Google Tasks writes.
- Real notifications.
- Real multi-user collaboration.

29. Portfolio Positioning

Product intuition statement

Gmail Commitment Radar is built on the insight that professional trust is often lost through small missed commitments, not because users lack writing tools. The prototype rethinks Gmail as a commitment-aware workspace that detects obligations, explains them transparently, and helps users resolve them inside the existing email workflow.

What this demonstrates

This project demonstrates:

- AI product thinking.
- Workflow automation.
- Gmail-native UI judgment.
- Trust and explainability.
- Professional productivity insight.
- Human override design.
- Data modeling.
- Risk scoring.
- Consumer-grade interaction polish.
- PM-level requirements depth.

Best resume or portfolio framing

“Designed and prototyped a Gmail-native AI commitment layer that detects promises, asks, deadlines, and follow-ups across email threads, then surfaces explainable, action-ready commitment cards with draft replies, snooze, task creation, and risk scoring.”

30. Build Prioritization

Phase 1: Static Gmail shell

Build:

- Top bar.
- Left nav.
- Inbox list.
- Thread view.
- Right panel.

Goal:

Make the prototype visually believable.

Phase 2: Seeded commitment data

Build:

- Mock threads.
- Commitment objects.
- Badges.
- Panel cards.
- Dashboard.

Goal:

Make the product concept clear.

Phase 3: Actions and state

Build:

- Mark done.
- Snooze.
- Dismiss.
- Draft reply.
- Edit due date.
- Snackbar undo.

Goal:

Make the product feel interactive.

Phase 4: AI simulation

Build:

- Rule-based analyzer.
- Confidence.
- Risk.
- Explanation.
- Source phrase highlighting.

Goal:

Make the intelligence feel grounded.

Phase 5: Polish

Build:

- Loading states.
- Empty states.
- Error states.
- Keyboard accessibility.
- Responsive behavior.
- Animation.

Goal:

Make it portfolio-ready.

31. Final MVP Definition

The MVP is complete when a user can open a Gmail-like inbox, immediately see which threads contain commitments, open a thread, understand exactly why a commitment was detected, take a lightweight action, and view all commitments in a Gmail-native dashboard.

The prototype should leave the viewer with one clear reaction:

“This feels like a feature Gmail should already have.”